Learning Objective 3.7: I can apply amplitude tapering (uniform, cosine, Chebyshev, Taylor) to control sidelobe level and predict the pattern trade-off.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Reading a taper off the sliders.** You find the eight Element Gains sliders on the PHASER set to

$$30,\ 60,\ 90,\ 100,\ 100,\ 90,\ 60,\ 30 \text{ percent.}$$

The array is the standard 8-element row, broadside, with the HB100 at boresight. A uniform-taper sweep has already been frozen as the reference trace.

(a) Compute the amplitude sum $\sum a_n$ and the peak drop this taper will show on the sweep plot relative to the frozen uniform trace.

$\sum a_n$ = 5.60

peak drop = -3.10 dB

The percentages are voltage amplitudes, so work in fractions of full scale:

$$\sum a_n = 2\,(0.30 + 0.60 + 0.90 + 1.00) = 5.60$$

The eight signals add in phase at boresight, so the summed voltage is proportional to $\sum a_n$ instead of $N = 8$:

$$\begin{aligned}\text{peak drop} &= 20\log_{10}(5.60/8)\\ &= 20\log_{10}(0.700) = -3.10 \text{ dB}\end{aligned}$$

(b) Compute the taper efficiency η_t for this distribution, as a ratio and in dB.

η_t = 0.867

η_t = -0.62 dB

$$\sum a_n^2 = 2\,(0.09 + 0.36 + 0.81 + 1.00) = 4.52$$

$$\eta_t = \frac{\left(\sum a_n\right)^2}{N\sum a_n^2} = \frac{5.60^2}{8 \times 4.52}$$

$$= \frac{31.36}{36.16} = 0.867 \quad (-0.62 \text{ dB})$$

(c) Your two answers differ by 2.5 dB. State which one the Rectangular plot shows you, and which one belongs in a link budget as the antenna's loss.

The plot shows the -3.10 dB peak drop. That is the coherent receive-voltage loss, and it appears only because every trace is referenced to the same uniform-taper full scale. The link budget takes the -0.62 dB taper efficiency, because that is the directivity the array gave up. Charging a link budget the full 3.10 dB overstates the antenna's loss by 2.5 dB.

(d) The uniform 8-element array has a broadside directivity of about 8.9 dB. What is the directivity of the tapered array?

D = 8.3 dBi

Directivity falls by the taper efficiency and by nothing else:

$$D = 8.9\,\text{dB} - 0.62\,\text{dB} = 8.3\,\text{dB}$$

The peak drop plays no part in this calculation.

2. **Predicting a sweep.** A lab group presses the **Blackman** preset, which sets the eight elements to

$$6,\ 27,\ 66,\ 100,\ 100,\ 66,\ 27,\ 6 \text{ percent},$$

and prepares to sweep. Their frozen uniform reference measured a 13.1° half-power beamwidth with its first sidelobe at -13 dBc, and the sweep's noise floor sits 23 dB below the uniform-taper peak.

(a) Predict the peak drop the Blackman trace will show relative to the frozen uniform trace.

peak drop = -6.1 dB

$$\sum a_n = 2\,(0.06 + 0.27 + 0.66 + 1.00) = 3.98$$
$$\text{peak drop} = 20\log_{10}(3.98/8)$$
$$= 20\log_{10}(0.4975) = -6.1 \text{ dB}$$

(b) The measured Blackman beamwidth is 23.1°. By what factor has the beam broadened, and what physical change in the array produced it?

broadening = $1.76\times$

$$\frac{23.1^\circ}{13.1^\circ} = 1.76$$

The taper leaves the end elements at 6% of full amplitude, so the outer aperture contributes almost nothing. The array's effective aperture is smaller than its physical length, and beamwidth is set by aperture size.

(c) Theory puts Blackman's first sidelobe roughly 33 dB below its own peak. What will the group see at that angle on the sweep, and what should they write in the sidelobe column of their table?

The Blackman peak sits 6.1 dB below the uniform peak, so a sidelobe 33 dB below it lands about 39 dB below full scale — roughly 16 dB below the noise floor. Nothing is visible at that angle except the floor itself. The correct entry is **below the noise floor**. Reading a number off the grass would report the receiver's noise, not the antenna's sidelobe.

(d) A group member argues that Blackman is the better taper because its sidelobes are lower than Hann's. Using only what this measurement can show, evaluate that claim.

This measurement cannot support the claim. Hann's sidelobes are already below the noise floor, so both tapers produce the same observable result — no visible sidelobes. Blackman costs 3.6° more beamwidth and 1.4 dB more peak drop than Hann, for an improvement the sweep cannot resolve. Against this receiver, Hann is the better choice; Blackman only earns its cost if the floor is lowered far enough to see Hann's sidelobes.

3. **A measurement that disagrees.** A student applies the **Hann** preset $(12,\ 43,\ 77,\ 100,\ 100,\ 77,\ 43,\ 12$ percent), predicts a -4.7 dB peak drop from the gain list, and measures -6.8 dB against the frozen uniform trace. The beam is also wider than the 19.5° the expectation table predicts.

   (a) Run the peak-drop formula backwards to find the amplitude sum the array achieved.

   $$\sum a_n = N \times 10^{-6.8/20} = 8 \times 0.4571 = 3.66$$

   $\sum a_n$ = 3.66

   The commanded distribution sums to 4.64, so the array is short by $4.64 - 3.66 = 0.98$.

   (b) Identify the fault. Which element is misbehaving, and what amplitude is it contributing?

   The shortfall of 0.98 is one full-amplitude element, so one of the two center elements (commanded to 100%) is contributing nothing — its gain is set to 0%, or its channel has failed. No combination of the outer elements can account for a deficit that large: the two end elements together only carry 0.24.

   (c) Compute the taper efficiency of the array as it stands, with that element at zero.

   The working elements are $0.12,\ 0.43,\ 0.77,\ 1.00,\ 0,\ 0.77,\ 0.43,\ 0.12$:

   $$\sum a_n = 3.64, \qquad \sum a_n^2 = 2.584$$

   $$\eta_t = \frac{3.64^2}{8 \times 2.584} = \frac{13.25}{20.67} = 0.641$$

   η_t = 0.641

   η_t = -1.9 dB

   which is -1.9 dB , against -1.2 dB for the intended Hann taper.

   (d) Name one feature of the trace itself, other than the peak level, that confirms your diagnosis.

   A dead center element breaks the symmetry of the distribution, so the measured pattern is no longer symmetric about boresight and the beam peak shifts slightly off 0°. The gap in the middle of the aperture also raises the sidelobes back out of the noise floor, which a correctly applied Hann taper had buried.

4. **Designing to a specification.** You need a taper that keeps the half-power beamwidth no wider than 17° while pushing the first sidelobe below -20 dBc. With Enforce Symmetric Taper on, you set a linear ramp from 40% at the ends to 91% at the center:

$$40,\ 57,\ 74,\ 91,\ 91,\ 74,\ 57,\ 40 \text{ percent.}$$

The measured sweep returns a 15.5° beamwidth with nothing visible above the noise floor outside the main lobe.

(a) Compute the peak drop and the taper efficiency of this design.

$$\sum a_n = 2\,(0.40 + 0.57 + 0.74 + 0.91) = 5.24$$
$$\sum a_n^2 = 2\,(0.160 + 0.325 + 0.548 + 0.828) = 3.721$$
$$\text{peak drop} = 20\log_{10}(5.24/8) = -3.7 \text{ dB}$$
$$\eta_t = \frac{5.24^2}{8 \times 3.721} = 0.922 \quad (-0.35 \text{ dB})$$

peak drop = -3.7 dB

η_t = 0.922

(b) The measurement shows nothing above the noise floor outside the main lobe. Explain what you can and cannot conclude about whether the -20 dBc part of the specification is met.

The floor sits 23 dB below the uniform peak, which is 19.3 dB below this trace's own peak. Nothing visible outside the main lobe therefore means the first sidelobe is at least 19.3 dB down. That demonstrates the specification is met to within the resolution of the measurement, but it does not produce a sidelobe number. To verify -20 dBc directly, the floor has to be lowered — a stronger source, more averaging, or a quieter receiver.

(c) Compare this design against the Hann preset (-4.7 dB peak drop, -1.2 dB taper efficiency, 19.5° beamwidth). State which you would fly on a radar that must resolve two targets 18° apart, and why.

The custom taper keeps 15.5° of beamwidth against Hann's 19.5°, and gives up only 0.35 dB of directivity against Hann's 1.2 dB. Two targets 18° apart fall inside a 19.5° beam and cannot be separated, so Hann fails the requirement outright. The custom taper resolves them and costs less directivity. Hann buys deeper sidelobes, but on this receiver that improvement is below the noise floor and cannot be measured.

beamwidth margin = 2.5°

(d) Enforce Symmetric Taper was on for this design. State what would change in the measured pattern if you set the same eight values with the switch off and mistyped one of them.

An asymmetric amplitude distribution produces an asymmetric pattern: the two sides of the trace no longer match, the sidelobes on one side rise above those on the other, and the beam peak moves off boresight. The symmetry switch makes that class of error impossible to enter by hand, which is why the lab keeps it on.

Documentation: _____